SQL PROJECT

NETFLIX

DATA ANALYSIS & INSIGHTS





A SQL based analysis of Netflix data to explore content trends, user preferences, ratings, cast & crew information and more through structured queries and meaningful insights.



```sql
SELECT title, type, release_year,
       rating, country
FROM netflix_data
WHERE type = 'Movie'
ORDER BY release_year DESC
LIMIT 10;
```

DATABASE DESIGN | DATA EXPLORATION | SQL QUERIES | INSIGHTS & ANALYSIS

PREPARED BY
SULTAN KHAN

Abstract

This project presents the design and implementation of a **Netflix Content Analysis Database Management System** using SQL. The system is developed to store, manage, and analyze information about Netflix titles, including movies and TV shows, their genres, release years, languages, ratings, directors, and countries of origin. The database consists of five core tables — Titles, Genres, Countries, Cast/Directors, and Ratings — connected through primary and foreign key relationships to ensure data integrity and efficient querying.

SQL KEYS AND FUNCTIONS

1.PRIMARY KEY -A column that uniquely identifies each row in a table. It cannot contain NULL or duplicate values.

2.FORIEGN KEY- A column that links one table to the Primary Key of another table.

3.CANDIDATE KEY- A column (or set of columns) that can uniquely identify a record. One candidate key becomes the Primary Key.

4. Alternate Key- Candidate keys that are not selected as the Primary Key.

5. Composite Key- A Primary Key made up of two or more columns.

6.UNIQUE KEY- Ensures all values are unique. Usually allows one NULL value (depends on DBMS).

FUNCTIONS IN SQL

1.AGGREGATE FUNCTION:

. COUNT-Counts rows

. SUM-Adds values

. AVG-Calculates average

.MAX-RETURNS HIGHEST VALUES

. MIN-RETURNS LOWEST VALUES

2.NUMERIC FUNCTION:

. ROUND-Rounds a number

. ABS-Returns absolute value

. CEIL-Rounds up

. FLOOR-Rounds down

3.STRING FUNCTION:

. UPPER-CONVERTS TEXT TO UPPERCASE

. LOWER-CONVERTS TEXT TO LOWERCASE

. LENGTH-RETURNS LENGTH OF TEXT

. CONCAT-JOINS TWO STRINGS

.TRIM-REMOVES SPACES

4.DATE FUNCTION:

. CURDATE-Current date

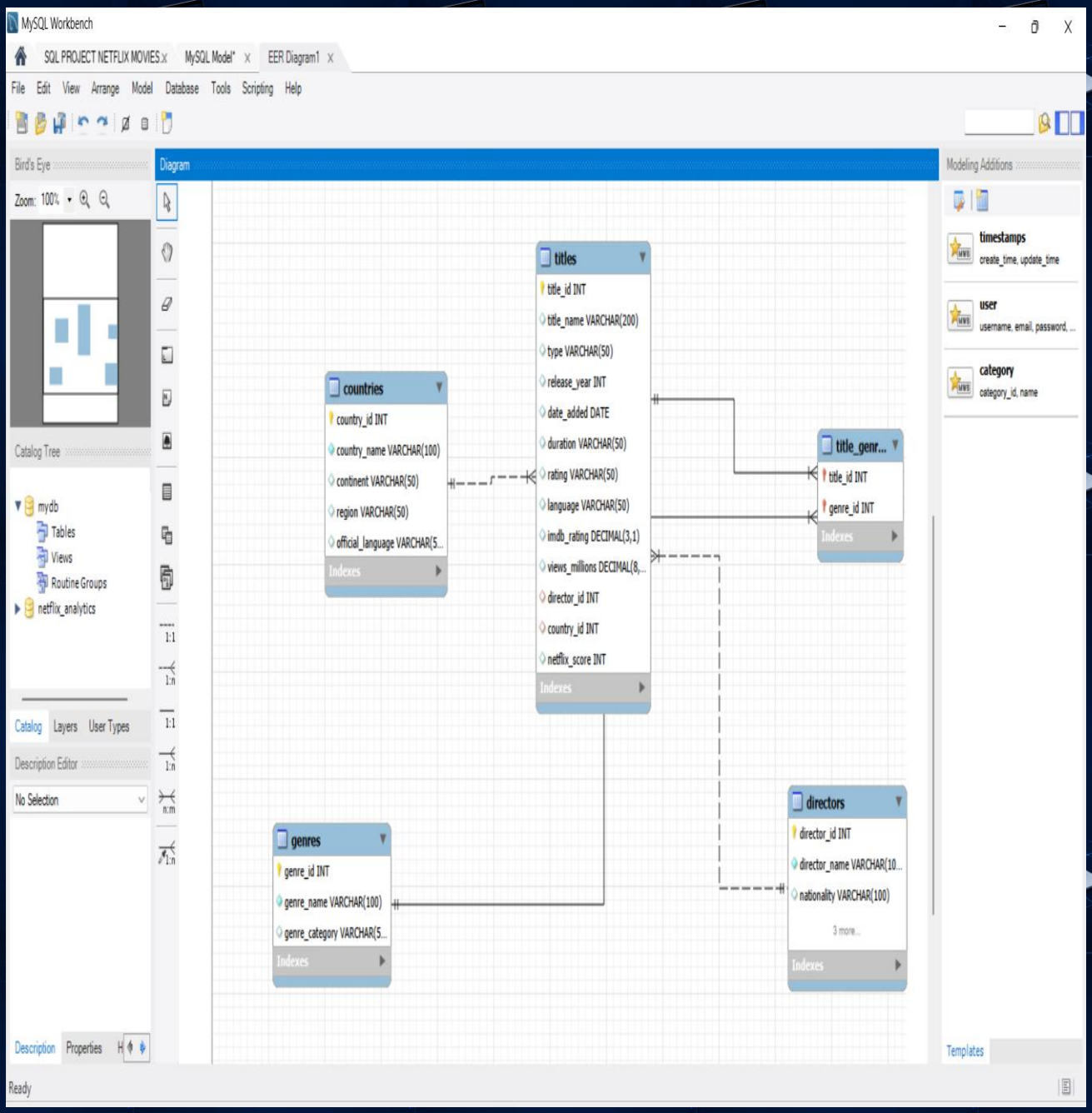
. NOW-CURRENT DATE AND TIME

. YEAR-EXTRACTS YEAR

SQL is divided into several categories of commands:

- **DDL (Data Definition Language):** Used to create and modify database objects (e.g., CREATE, ALTER, DROP).
- **DML (Data Manipulation Language):** Used to insert, update, and delete data (e.g., INSERT, UPDATE, DELETE).
- **DQL (Data Query Language):** Used to retrieve data from the database (e.g., SELECT).
- **DCL (Data Control Language):** Used to manage user permissions (e.g., GRANT, REVOKE).
- **TCL (Transaction Control Language):** Used to control transactions (e.g., COMMIT, ROLLBACK, SAVEPOINT).

ER DIAGRAM:


MySQL Workbench
SQL PROJECT NETFLIX MOVIES
MySQL Model*
EER Diagram1
File Edit View Arrange Model Database Tools Scripting Help
Bird's Eye
Zoom: 100%
Catalog Tree
mydb
Tables
Views
Routine Groups
netflix_analytics
Catalog Layers User Types
Description Editor
No Selection
Description Properties
Diagram
titles
title_id INT
title_name VARCHAR(200)
type VARCHAR(50)
release_year INT
date_added DATE
duration VARCHAR(50)
rating VARCHAR(50)
language VARCHAR(50)
imdb_rating DECIMAL(3,1)
views_millions DECIMAL(8,...
director_id INT
country_id INT
netflix_score INT
Indexes
countries
country_id INT
country_name VARCHAR(100)
continent VARCHAR(50)
region VARCHAR(50)
official_language VARCHAR(5...
Indexes
title_genr...
title_id INT
genre_id INT
Indexes
genres
genre_id INT
genre_name VARCHAR(100)
genre_category VARCHAR(5...
Indexes
directors
director_id INT
director_name VARCHAR(10...
nationality VARCHAR(100)
3 more...
Indexes
Modeling Additions
timestamps
create_time, update_time
user
username, email, password, ...
category
category_id, name
Templates
Ready

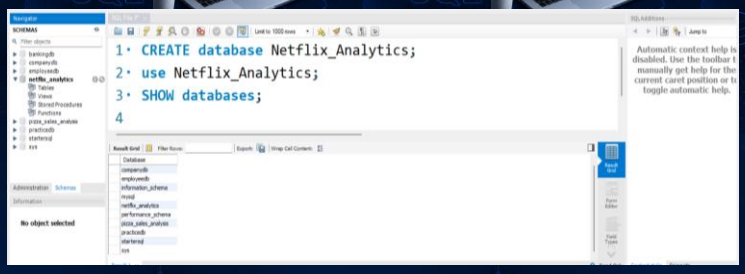
Database Explanation

The **Netflix Analytics database** is designed to store and analyze information about movies and TV shows available on Netflix. It follows a relational model with multiple interconnected tables:

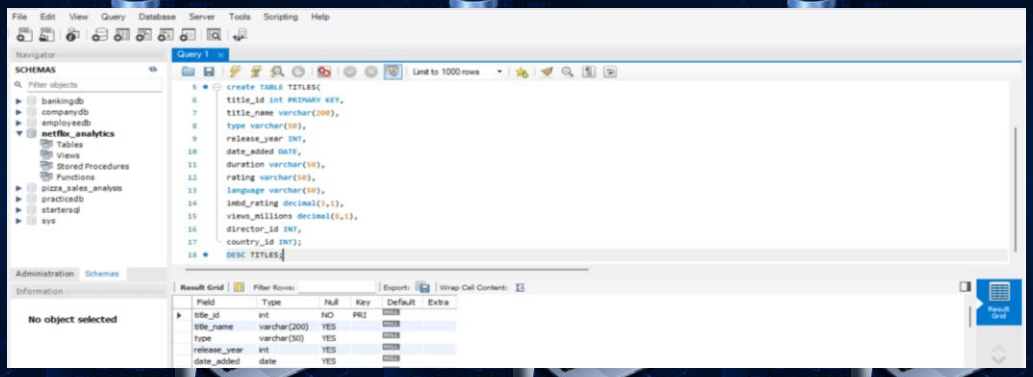
- **Titles**: Central table containing details of each movie or show (name, type, release year, duration, rating, language, IMDb rating, views, etc.).
- **Directors**: Stores information about directors, including nationality, birth year, awards, and career timeline.
- **Countries**: Represents the origin of titles, with continent, region, and official language.
- **Genres**: Contains categories such as Action, Drama, Comedy, Sci-Fi, etc.
- **Title_Genres**: A bridge table that links titles to multiple genres, enabling many-to-many relationships.

DATA DEFINTION LANGUAGE (DDL)-

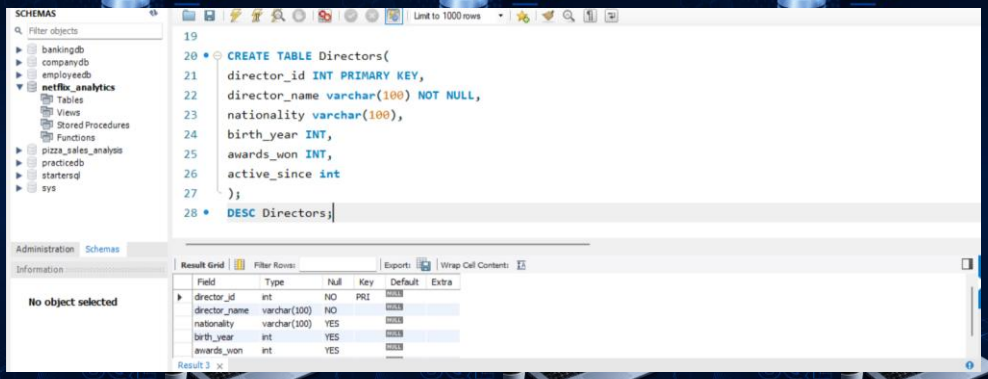
Create database:



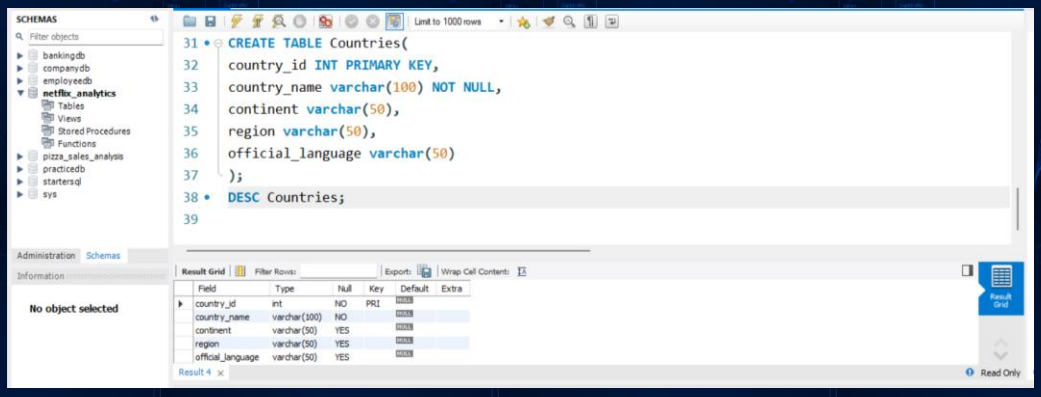
1.Create TABLE TITLES:



2.Create TABLE Directors:



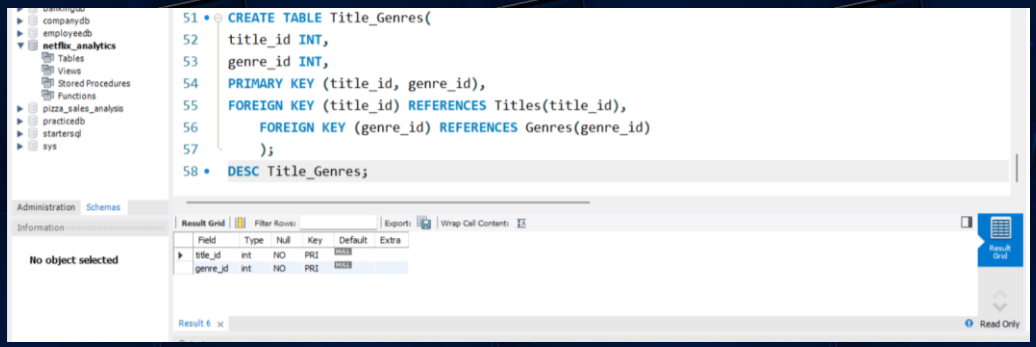
3.Create TABLE Countries:



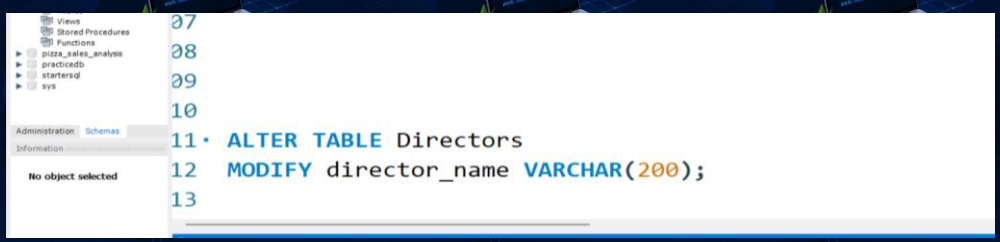
4.Create TABLE Genres:



5.Create TABLE Title_Genres:



6. Modify Column in Directors Table

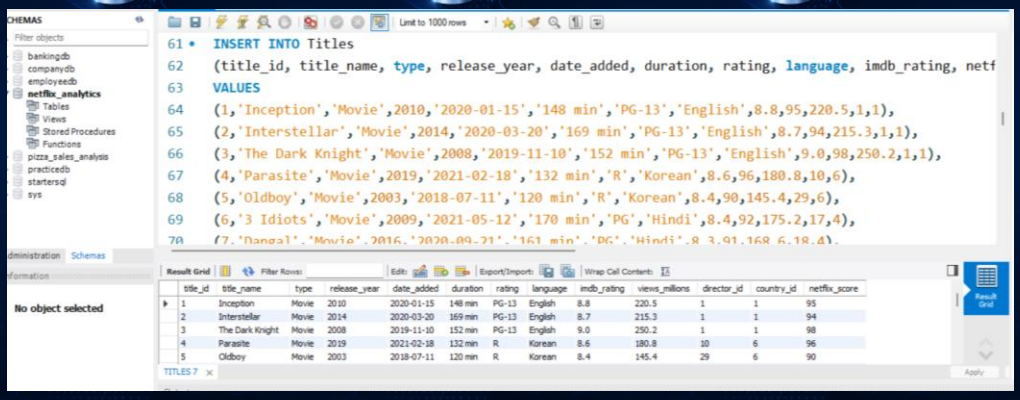


7. Add a Column to Titles Table

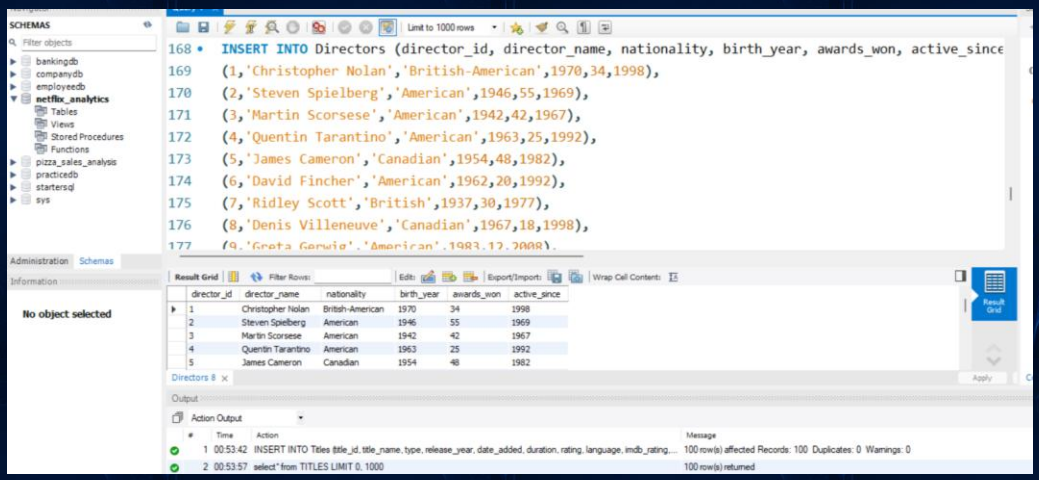


DATA MANUPULATION LANGUAGE (DML)-

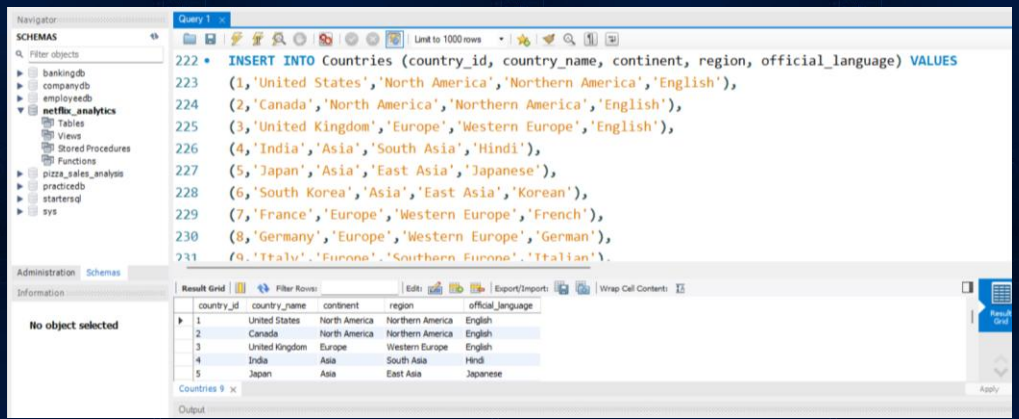
1.



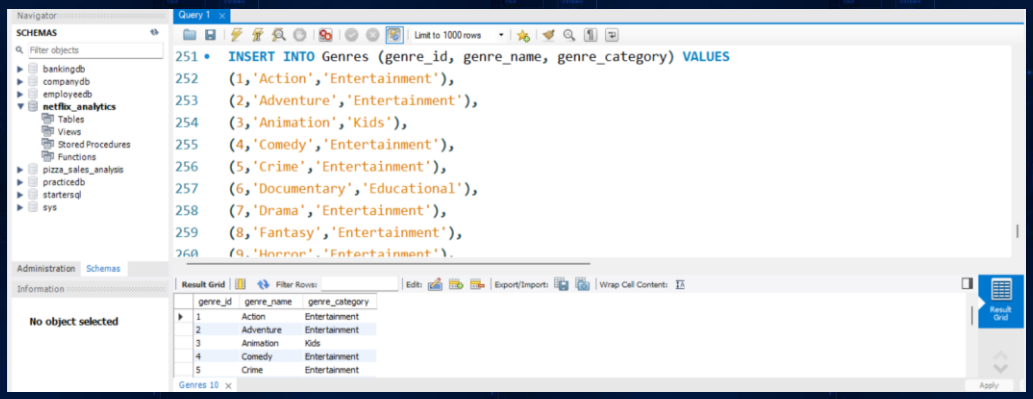
2.



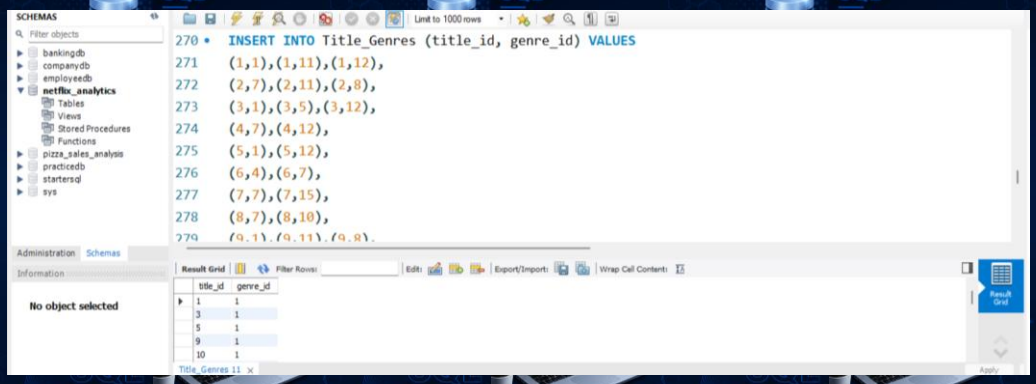
3.



4.


INSERT INTO Genres (genre_id, genre_name, genre_category) VALUES
(1,'Action','Entertainment'),
(2,'Adventure','Entertainment'),
(3,'Animation','Kids'),
(4,'Comedy','Entertainment'),
(5,'Crime','Entertainment'),
(6,'Documentary','Educational'),
(7,'Drama','Entertainment'),
(8,'Fantasy','Entertainment'),
genre_id genre_name genre_category
1 Action Entertainment
2 Adventure Entertainment
3 Animation Kids
4 Comedy Entertainment
5 Crime Entertainment
No object selected

5.


INSERT INTO Title_Genres (title_id, genre_id) VALUES
(1,1),(1,11),(1,12),
(2,7),(2,11),(2,8),
(3,1),(3,5),(3,12),
(4,7),(4,12),
(5,1),(5,12),
(6,4),(6,7),
(7,7),(7,15),
(8,7),(8,10),
title_id genre_id
1 1
3 1
5 1
9 1
10 1
No object selected

DELETE Queries (DML)

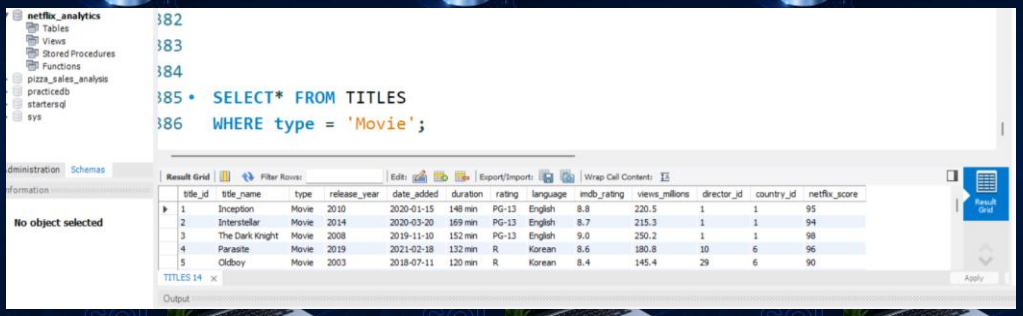
```
-- Delete titles released before 2000
DELETE FROM Titles
WHERE release_year < 2000;

-- Delete directors with no movies
DELETE FROM Directors
WHERE director_id NOT IN (SELECT director_id
FROM Titles);

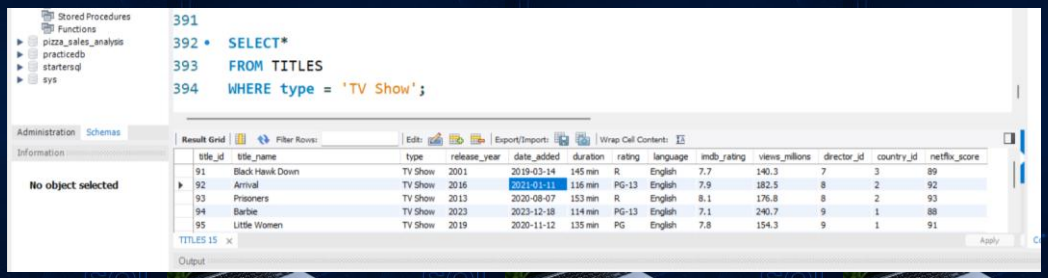
-- Delete genres not linked to any title
DELETE FROM Genres
WHERE genre_id NOT IN (SELECT genre_id
FROM Title_Genres);
```

BASIC QUERIES :

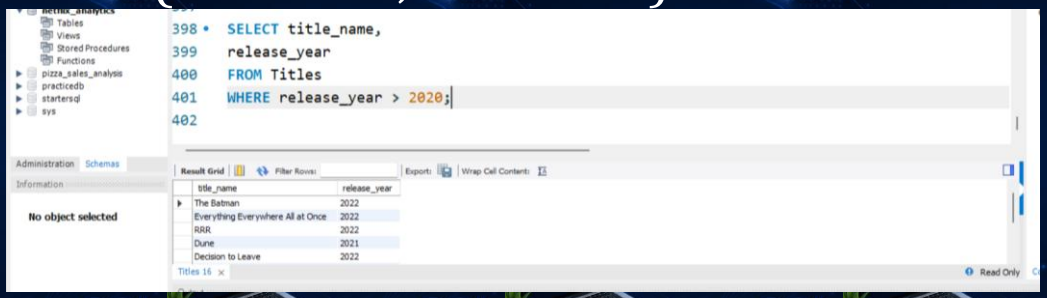
1.DISPLAY ALL MOVIES(statement used select,where)



2.DISPLAY ALL TV Shows

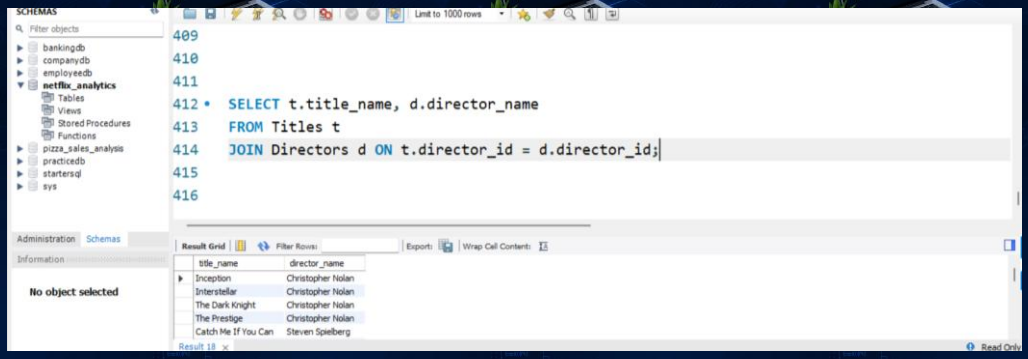


3.DISPLAY MOVIES WHICH RELEASED AFTER 2020(selection,condition)

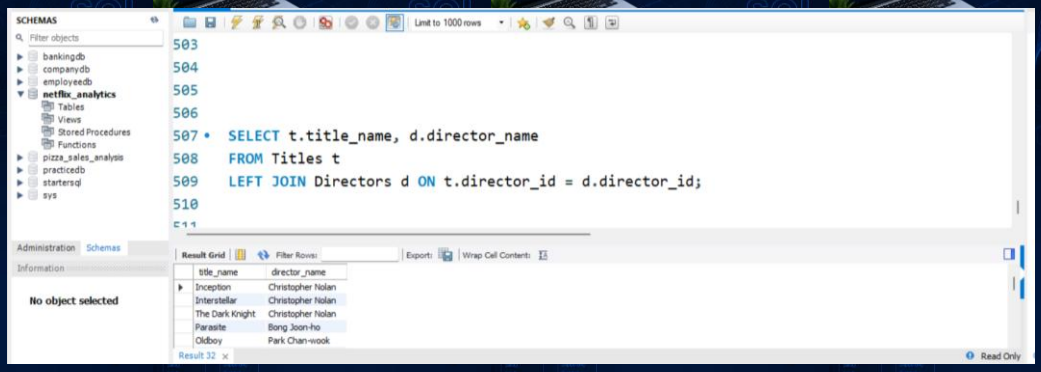


JOIN QUERIES

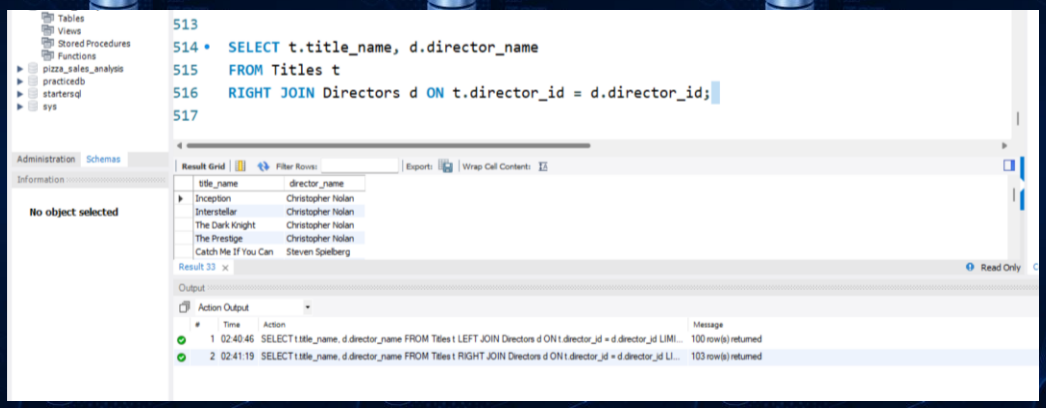
1.INNER JOIN(TITLES WITH DIRECTORS)



2.LEFT JOIN



3.RIGHT JOIN



Aggregation Functions

1.count (count total tiles)

```
SELECT COUNT(*) AS Total_Titles
FROM Titles;
```

Total_Titles
100

2.AVG Function:

```
SELECT AVG(imdb_rating) AS Avg_Rating
FROM Titles;
```

Avg_Rating
7.94800

3.GROUP BY:

```
SELECT c.country_name, COUNT(t.title_id) AS Total_Titles
FROM Titles t
JOIN Countries c ON t.country_id = c.country_id
GROUP BY c.country_name;
```

country_name	Total_Titles
United States	54
Canada	7
United Kingdom	10
India	4
Japan	6

4.HAVING CLAUSE:

```
SELECT d.director_name, COUNT(t.title_id) AS Total_Titles
FROM Directors d
JOIN Titles t ON d.director_id = t.director_id
GROUP BY d.director_name
HAVING COUNT(t.title_id) > 3;
```

director_name	Total_Titles
Christopher Nolan	4
Steven Spielberg	4
Martin Scorsese	5
Quentin Tarantino	5
David Fincher	7

STRING AND DATE FUNCTIONS

1.UPPER

```
460
461 •  SELECT UPPER(title_name) AS Title_UpperCase
462    FROM Titles
463    LIMIT 10;
464
465
```

Title_UpperCase
INCEPTION
INTERSTELLAR
THE DARK KNIGHT
PARASITE
OLDBOY

2.LIKE FUNCTION

```
464
465
466
467 •  SELECT title_name
468    FROM Titles
469    WHERE title_name LIKE '%King%';
470
471
```

title_name
King Kong
The Lion King

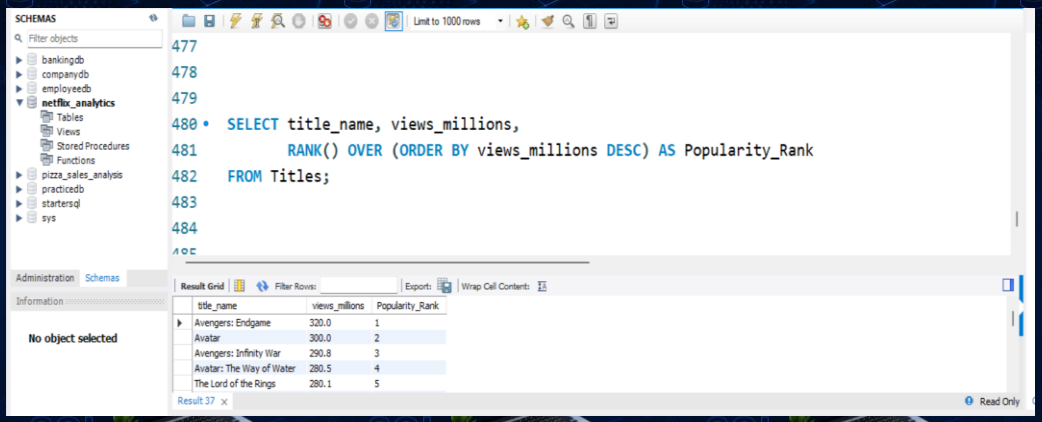
3.Date Function (YEAR, ORDER BY)

```
472
473 •  SELECT title_name, YEAR(date_added) AS Added_Year
474    FROM Titles
475    ORDER BY Added_Year DESC;
476
477
```

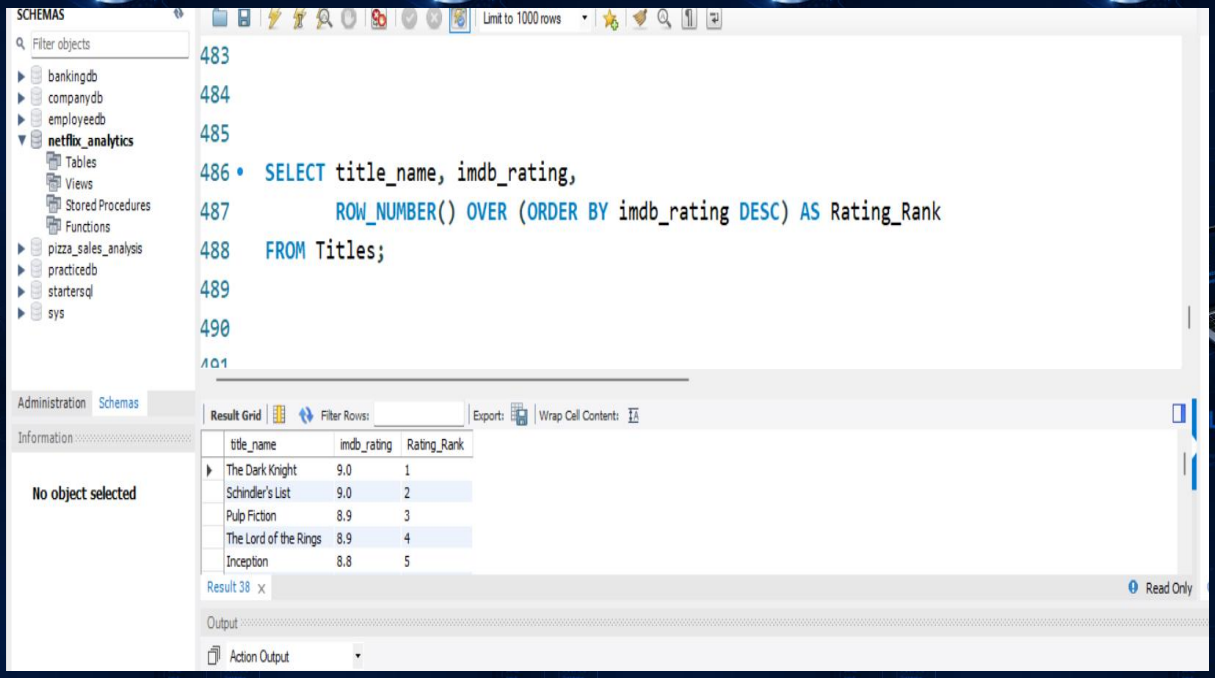
title_name	Added_Year
Elvis	2023
Barbie	2023
Avatar: The Way of Water	2023
Decision to Leave	2023
Everything Everywhere All at Once	2023

WINDOWS FUNCTIONS

1. RANK Function



2. ROW_NUMBER Function

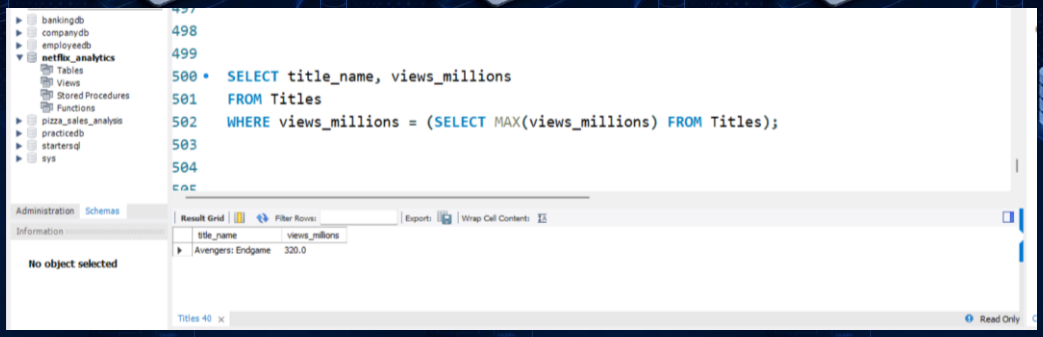


SUB QUERIES

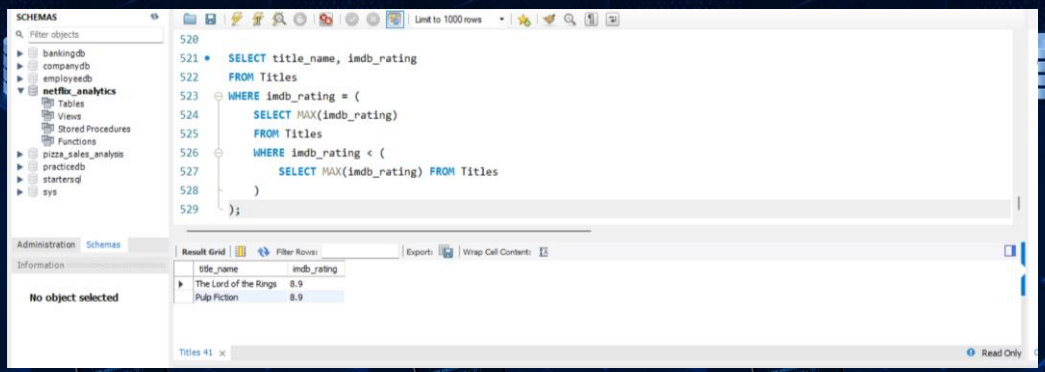
1. Subquery (Above Average Rating)



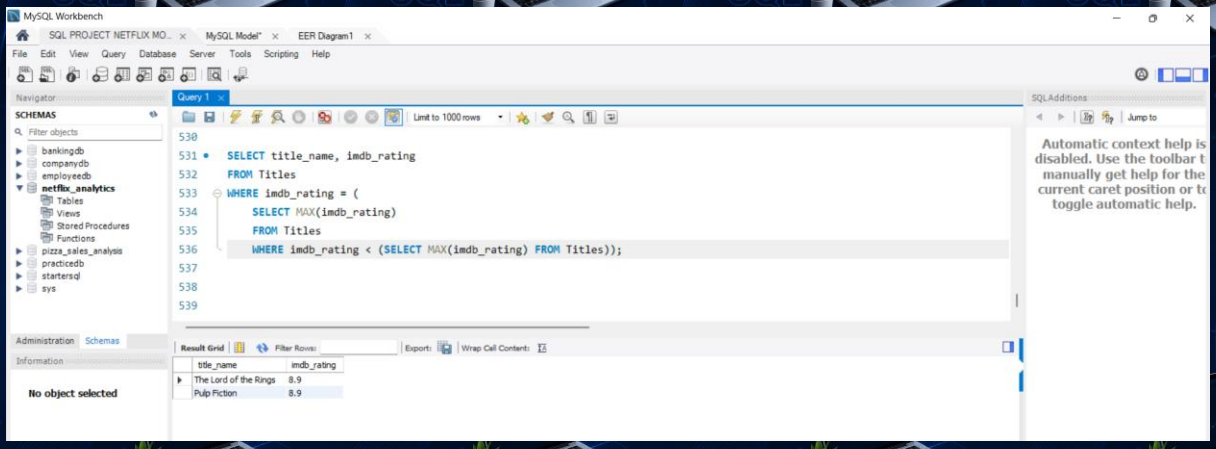
2. Subquery (MAX Value)



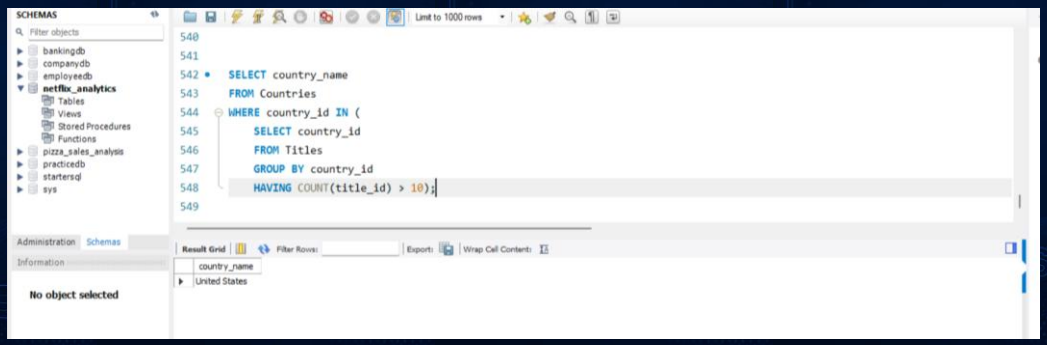
3. Subquery, MAX, WHERE



4.NESTED SUBQUERY



5. Subquery, GROUP BY, HAVING



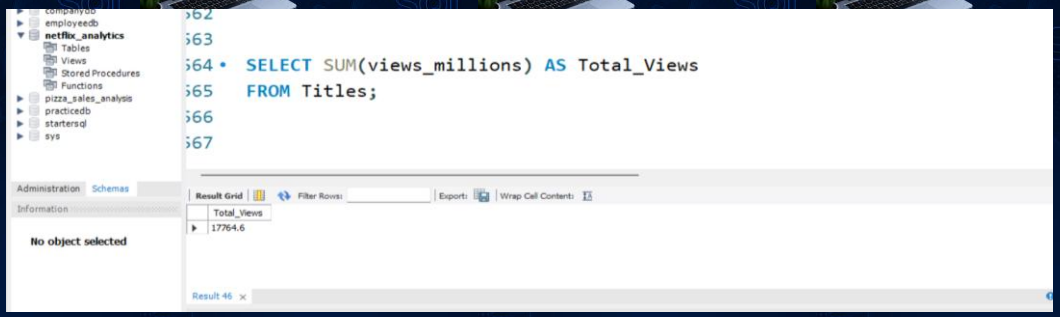
1. Maximum IMDb Rating (MAX)



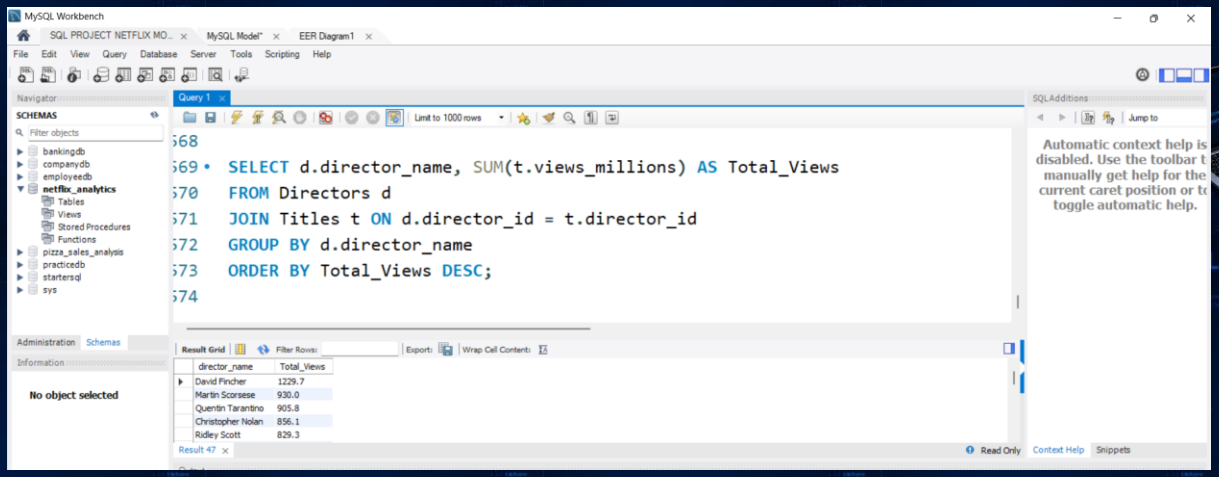
2. Minimum IMDb Rating (MIN)



4. Total Views of All Titles (SUM)



5. Total Views per Director (SUM with GROUP BY)



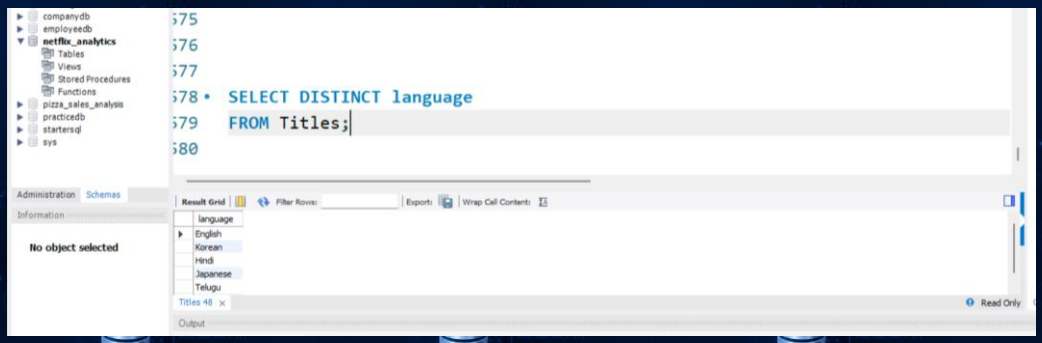
DISTINCT FUNCTION

The DISTINCT keyword removes duplicates.

This query shows all unique languages available in the Netflix dataset.

Useful for understanding diversity of content.

1. Distinct Languages (DISTINCT)



2. Distinct Directors (DISTINCT)



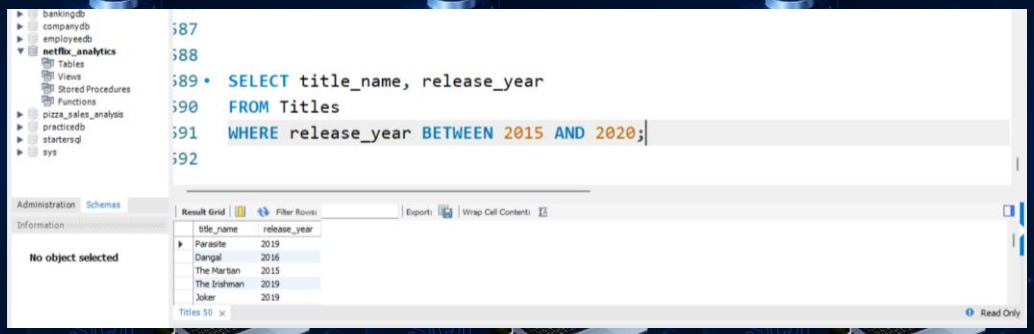
BETWEEN Function

The BETWEEN operator filters values within a range.

This query returns all titles released between 2015 and 2020.

Useful for analyzing content trends in a specific time period

1. Titles Released Between 2015 and 2020

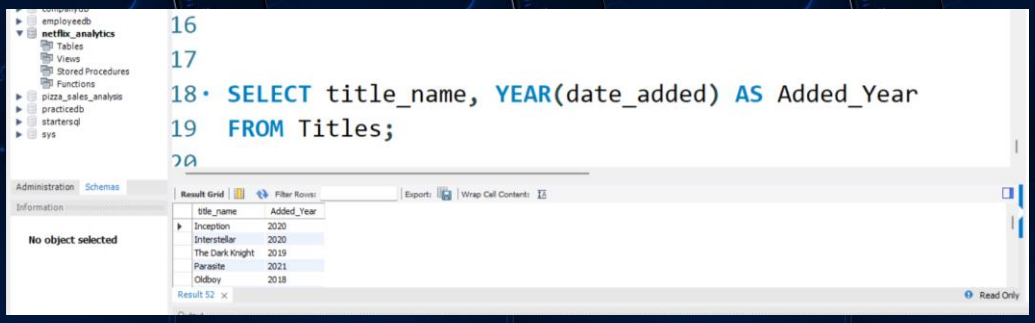


2. Titles with IMDb Rating Between 7 and 9 (BETWEEN)

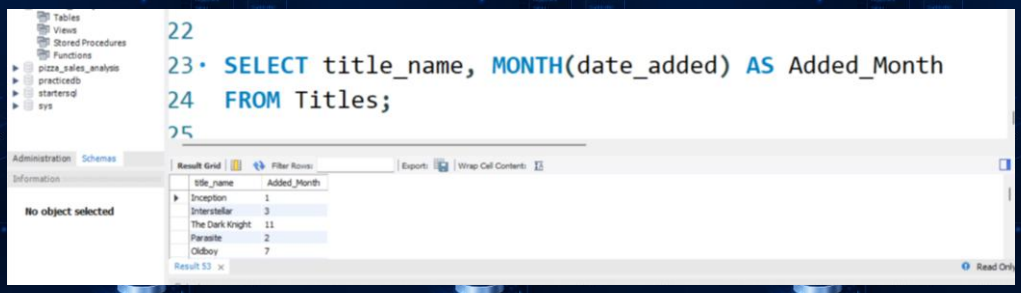


DATE FUNCTIONS

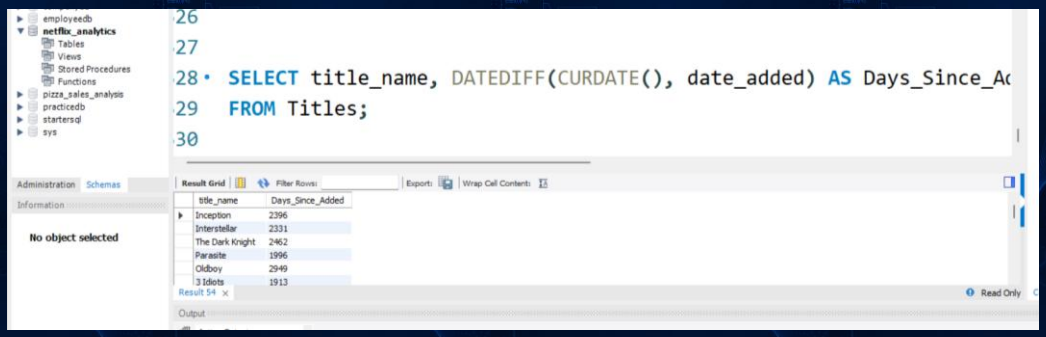
1.Extract Year from Date (YEAR)



2. Extract Month from Date (MONTH)

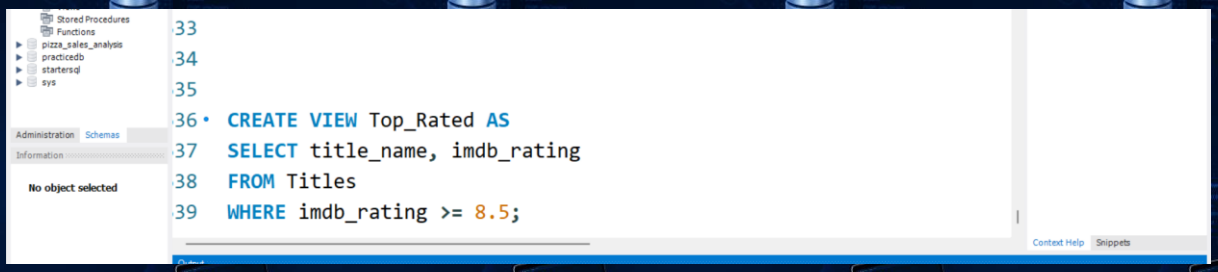


3. Difference in Days (DATEDIFF)

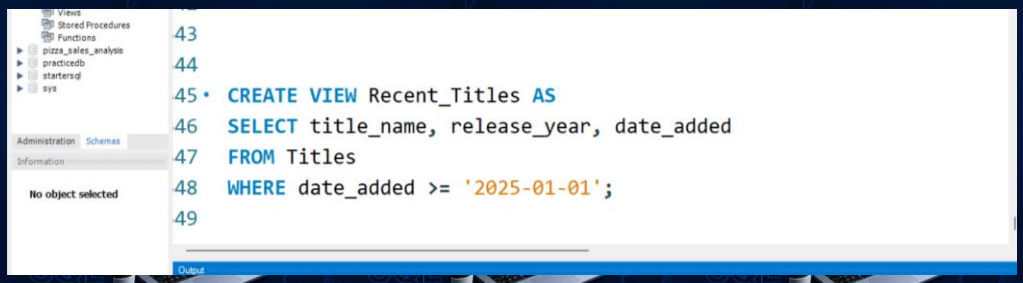


VIEWS FUNCTION

1. Create a view for top rated movies



2.Create a View for Recently Added Titles



CONCLUSION

In this project, we successfully explored the use of SQL for data analytics by applying a wide range of queries and functions. Starting from DDL operations (CREATE, ALTER, DROP) to define the database schema, we moved into DML operations (INSERT, UPDATE, DELETE, SELECT) to manipulate and analyze data.

We demonstrated the use of Joins to combine multiple tables, Aggregate functions to summarize information, and Filtering techniques to refine results. Advanced topics such as Window functions, String functions, and Date functions were applied to perform deeper analytics. Finally, the creation of Views showed how complex queries can be simplified for repeated use.

This comprehensive coverage highlights SQL's versatility as a powerful tool for managing and analyzing structured data. The project not only strengthened technical skills but also provided practical insights into how SQL supports real-world analytics tasks.